

Nonoverlapping functions of DNA polymerases mu, lambda, and terminal deoxynucleotidyltransferase during immunoglobulin V(D)J recombination in vivo.

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DNA polymerases mu (pol mu), lambda (pol lambda), and terminal deoxynucleotidyltransferase (TdT) are enzymes of the pol X family that share homology in sequence and functional domain organization. We showed previously that pol mu participates in light chain but surprisingly not heavy chain gene rearrangement. We show here that immunoglobulin heavy chain junctions from pol lambda-deficient animals have shorter length with normal N-additions, thus indicating that pol lambda is recruited during heavy chain rearrangement at a step that precedes the action of TdT. In contrast to previous in vitro studies, analysis of animals with combined inactivation of these enzymes revealed no overlapping or compensatory activities for V(D)J recombination between pol mu, pol lambda, and TdT. This complex usage of polymerases with distinct catalytic specificities may correspond to the specific function that the third hypervariable region assumes for each immunoglobulin chain, with pol lambda maintaining a large heavy chain junctional heterogeneity and pol mu ensuring a restricted light chain junctional variability.